

SIH Local AI Workbench: Inspection Report

Read-only repository and environment inventory, inspected 4 September 2026.

Persistence model

This project does not use SQLite, PostgreSQL, MySQL, or an ORM. The only persistent application data is a JSON account store in LocalAppData. Sessions, chat history, document metadata, extracted page text, and upload records are memory-only.

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Pinned direct dependencies	Persisted user accounts	Project-local persisted records	Document context cap (characters)

Installed Runtime

Area	Observed state
Python environment	Python: 3.14.3 in the project virtual environment. Ollama executable: installed at the per-user Program Files location, but not available on this shell PATH. Tesseract: installed at the standard 64-bit Program Files location, but not available on this shell PATH. The OCR code can locate Tesseract from its configured standard paths even when PATH is missing it.
Runtime topology	Desktop: Tkinter client in app.py, directly calling local Ollama and document modules. Web: FastAPI server in server/main.py, served by Uvicorn with a static browser client. Model default: llama3.2:latest, configurable through SIH_MODEL_NAME.

Declared Direct Dependencies

Package	Pinned version	Current use
ollama	0.6.2	Local Ollama client and streaming chat
fastapi	0.141.1	LAN/web API
uvicorn[standard]	0.52.4	ASGI server
pymupdf	1.28.2	PDF parsing and page rendering
pytesseract	0.3.13	OCR bridge to Tesseract
pillow	12.3.0	Pixmaps to OCR images
pydantic	2.13.5	HTTP request validation
python-multipart	0.0.32	Multipart PDF uploads
psutil	7.2.2	Host metrics
httpx2	2.12.0	FastAPI test client transport

Full Virtual Environment Inventory — 56 packages

Package	Version	Package	Version
annotated-doc	0.0.5	pip-requirements-parser	32.0.1
annotated-types	0.8.0	pip_audit	2.10.1
anyio	4.14.2	platformdirs	4.11.7
boolean.py	5.0	psutil	7.2.2
CacheControl	0.14.4	py-serializable	2.1.0
certifi	2026.7.22	pydantic	2.13.5
charset-normalizer	3.5.1	pydantic_core	2.46.5

Package	Version	Package	Version
click	8.5.0	Pygments	2.21.0
cyclonedx-python-lib	11.12.0	pymupdf	1.28.2
defusedxml	0.7.1	pyparsing	3.3.2
fastapi	0.141.1	pytesseract	0.3.13
filelock	3.32.5	python-dotenv	1.2.3
h11	0.16.0	python-multipart	0.0.32
httpcore	1.0.9	PyYAML	6.0.3
httpcore2	2.12.0	reportlab	5.0.1
httpptools	0.8.0	requests	2.34.2
httpx	0.28.1	rich	15.0.0
httpx2	2.12.0	sortedcontainers	2.4.0
idna	3.19	starlette	1.6.0
license-expression	30.4.4	tomli	2.4.1
markdown-it-py	4.2.0	tomli_w	1.2.0
mdurl	0.1.2	truststore	0.10.4
msgpack	1.2.2	typing-inspection	0.4.4
ollama	0.6.2	typing_extensions	4.16.0
packageurl-python	0.17.6	urllib3	2.7.0
packaging	26.3	uvicorn	0.52.4
pillow	12.3.0	watchfiles	1.2.0
pip-api	0.0.34	websockets	17.1

The virtual environment also contains transitive packages plus pip-audit and its audit dependencies.

Account Store Schema

Layer	Current structure
File	%LOCALAPPDATA%\SIHLocalAI\users.json
Format	JSON object with one users array
User fields	id, username, password_hash, role, is_active, created_at, last_login
Current records	2 users: one admin role and one user role
Password storage	scrypt N=16384, r=8, p=1; legacy PBKDF2 accepted once and upgraded at login
Write behavior	Locked, temp-file write, fsync, atomic replace

Volatile Session Schema

Object	In-memory fields
AuthToken	token, user_id, username, role, expires_at, session_id
SessionData	identity, role, timestamps, mode, separate general/document histories
DocumentMetadata	document/session IDs, safe filename, path, extraction and OCR metrics
DocumentExtraction	page count, page blocks, character count, timing, native/OCR stats, failures
DocumentContext	page-labelled text, included/total pages, truncation flag, estimated tokens

All session data is lost on server restart. The project-local data and uploads directories are currently empty.

Document-Processing Pipeline

Stage	Component	Observed behavior
1. Ingest	Browser upload or desktop file picker	PDF only; 25 MB and 80-page limits in the server flow
2. Isolate	FileHandler	Sanitized filename; per-session upload directory; PDF magic-byte check
3. Parse	PyMuPDF	Reads a bounded PDF byte stream; rejects encrypted, unreadable, or oversized page-count inputs
4. Extract	Native text first	Whitespace normalization; usable threshold: 30 chars and 12 alphanumeric characters
5. OCR fallback	Tesseract via pytesseract	220 DPI target, capped at 20 million pixels; English, page segmentation mode 6
6. Bound context	DocumentContext	Page-labelled context capped at 14,000 characters; estimate uses 4 chars per token
7. Ask model	Ollama llama3.2:latest	Document text is embedded as untrusted reference evidence; browser output is SSE streamed
8. Retain and clean	In-memory session + uploads	Chat history is bounded; uploads are deleted on clear, logout, or user removal

Chat and Document Routing

General chat sends the user message to the selected local model. Document chat first selects an owned document, rebuilds extraction/context when switching documents, applies the document-evidence prompt, and stores separate document-mode history. The web server permits one active inference at a time, streams browser output as server-sent events, caps generated output at 64,000 characters, and retains at most 40 non-system history messages with a 48,000-character history budget.

Source inspected: requirements.txt, app.py, run_server.py, core modules, document modules, server/main.py, project state directories, and the active project virtual environment. No project files, account records, documents, or settings were changed.